The Solutions

B—bookwork, A—application, E—new example, T—new theory

1. Random variables.

a)

i) Using normalization $\iint f(x,y)dxdy = 1$, we have $c_1 = 6$. [2A]

$$f(x) = 2x, \ \ 0 < x < 1$$ [1A]

$$f(y) = 3y^2, \ \ 0 < y < 1$$ [1A]

Independent because $f(x,y) = f(x)f(y)$. [1A]

ii) $c_2 = 3$ [2A]

$$f(x) = x + \frac{3}{2}x^2, \ \ 0 < x < 1$$ [1A]

$$f(y) = y + \frac{3}{2}y^2, \ \ 0 < y < 1$$ [1A]

Dependent because $f(x,y) \neq f(x)f(y)$. [1A]

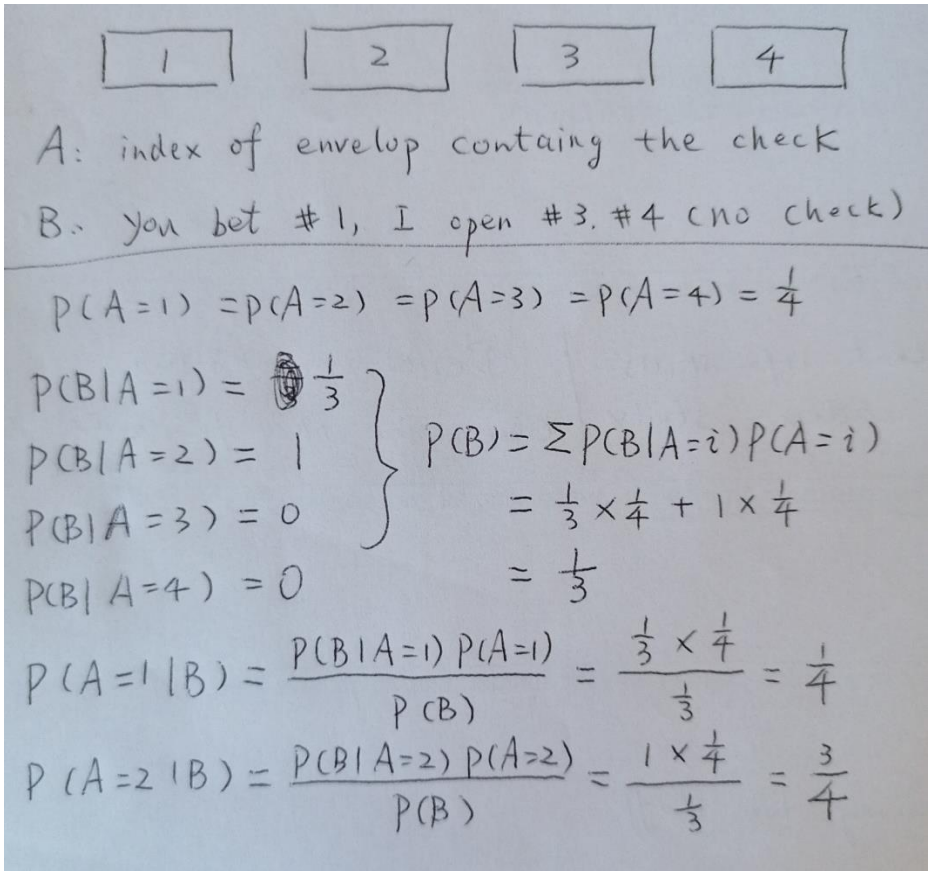
iii) $c_3 = \frac{1}{\sqrt{\pi}}$ [2A]

$$f(x) = e^{-x}, \quad 0 < x < \infty.$$ [1A]

$$f(y) = \frac{1}{\sqrt{\pi}} e^{-y^2}, \quad -\infty < y < \infty.$$ [1A]

Independent because $f(x,y) = f(x)f(y)$. [1A]

b)



[2E]

[2E]

[2E]

[2E]

[2E]

Although the a priori distribution is uniform, the a posteriori probabilities are different. The probability of Envelope 2 has increased from 1/4 to 3/4. So you should switch.

2. Estimation and sequences of random variables.

a) For this problem, it's easier to re-label σ^2 by another variable, say, θ.

i) Then [1E]

$$f(x) = \frac{x}{\theta} e^{-\frac{x^2}{2\theta}}, \qquad x > 0.$$ [2E]

Then,

$$f(\boldsymbol{x}, \theta) = \frac{(x_1 \dots x_n)}{\theta^n} e^{-\frac{(x_1^2 + \cdots + x_n^2)}{2\theta}}$$

The log-likelihood function reads [2E]

$$\ln f(\boldsymbol{x}, \theta) = \ln(x_1 \dots x_n) - n \ln \theta - \frac{(x_1^2 + \cdots + x_n^2)}{2\theta}$$

$$\frac{\partial \ln f(\boldsymbol{x}, \theta)}{\partial \theta} = -\frac{n}{\theta} + \frac{(x_1^2 + \cdots + x_n^2)}{2\theta^2} = 0$$

$$\hat{\theta} = \frac{(x_1^2 + \cdots + x_n^2)}{2n}$$

ii) In this problem, n = 5. [2E]

$$\hat{\theta} = \frac{(3^2 + 4^2 \dots + 7^2)}{10} = 2.3.$$

iii) Unbiased, because [3E]

$$E[\hat{\theta}] = \frac{E(x_1^2 + \cdots + x_n^2)}{2n} = \frac{n \cdot 2\sigma^2}{2n} = \sigma^2$$

b) Let $\mu = E[X]$.

i) Markov

$$P(X > a) \leq \frac{\mu}{a} = \frac{\sqrt{\frac{\pi}{2}}\sigma}{a} = \frac{\sqrt{\frac{\pi}{2}}\sigma}{3\sigma} = 0.418$$ [5A]

ii) Chebyshev

$$P(X > a) = P(X - \mu > a - \mu) = P(|X - \mu| > a - \mu)$$

$$\leq \frac{Var(X)}{(a-\mu)^2} = \frac{\frac{4-\pi}{2}\sigma^2}{\left(3 - \sqrt{\frac{\pi}{2}}\right)^2 \sigma^2} = 0.141$$ [5A]

iii) Exact

Since $F(x) = 1 - e^{-\frac{x^2}{2\sigma^2}}$, [5A]

$$P(X > a) = e^{-a^2/(2\sigma^2)} = e^{-4.5} = 0.0222$$

3. Random processes.

a)

i) It's clear that $E[X(n)] = 0$. [2E]

The autocorrelation function

$$R_X(m, m+n) = E[(A\cos(n\lambda) + B\sin(n\lambda))(A\cos((n+m)\lambda) + B\sin((n+m)\lambda))] \quad [2E]$$

$$= \cos(n\lambda)\cos((n+m)\lambda) + \sin(n\lambda)\sin((n+m)\lambda)$$

$$+ \frac{1}{2}[\cos(n\lambda)\sin((n+m)\lambda) + \sin(n\lambda))\cos((n+m)\lambda)] \quad [3E]$$

$$= \cos(m\lambda) + \frac{1}{2}\sin((2n+m)\lambda)$$

ii) Obviously, it is not stationary, since the autocorrelation function depends on n. [3E]

b)

i) Using Cauchy-Schwarz inequality,

$$SNR = \frac{\left|\frac{1}{2\pi}\int_{-\infty}^{+\infty} S(\omega)H(\omega)e^{j\omega t_0}d\omega\right|^2}{\frac{1}{2\pi}\int_{-\infty}^{+\infty} S_w(\omega)|H(\omega)|^2\,d\omega}$$

$$= \frac{\left|\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{S(\omega)}{\sqrt{S_w(\omega)}}\sqrt{S_w(\omega)}H(\omega)e^{j\omega t_0}d\omega\right|^2}{\frac{1}{2\pi}\int_{-\infty}^{+\infty} S_w(\omega)|H(\omega)|^2\,d\omega} \quad [2T]$$

$$\le \frac{1}{2\pi}\frac{\int_{-\infty}^{+\infty}\frac{S^2(\omega)}{S_w(\omega)}d\omega\int_{-\infty}^{+\infty} S_w(\omega)|H(\omega)e^{j\omega t_0}|^2\,d\omega}{\int_{-\infty}^{+\infty} S_w(\omega)|H(\omega)|^2\,d\omega} \quad [2T]$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{S^2(\omega)}{S_w(\omega)}d\omega$$

where equality holds if and only if

$$\frac{S^*(\omega)}{\sqrt{S_w(\omega)}} = k\sqrt{S_w(\omega)}H(\omega)e^{j\omega t_0}, \quad k \text{ a constant, can be ignored}$$

$$H(\omega) = \frac{S^*(\omega)}{S_w(\omega)}e^{-j\omega t_0} \quad [2T]$$

Since the Fourier transform of the signal is still Gaussian

$$S(\omega) = e^{-\sigma_s^2\omega^2/2}$$

we have

$$H(\omega) = \frac{e^{-\sigma_s^2\omega^2/2}}{N_0 e^{-\sigma_w^2\omega^2/2}} e^{-j\omega t_0} = \frac{1}{N_0} e^{-(\sigma_s^2-\sigma_w^2)\omega^2/2} e^{-j\omega t_0}$$ [2T]

The matched filter is still Gaussian

$$h(t) = \frac{1}{N_0\sqrt{2\pi(\sigma_s^2-\sigma_w^2)}} e^{-\frac{(t_0-t)^2}{2(\sigma_s^2-\sigma_w^2)}}$$ [3T]

ii) maximum SNR

$$SNR = \frac{1}{2\pi}\int \frac{S^2(\omega)}{S_W(\omega)} d\omega$$

$$= \frac{1}{2\pi}\int \frac{e^{-\sigma_s^2\omega^2}}{N_0 e^{-\frac{\sigma_w^2\omega^2}{2}}} d\omega = \frac{1}{N_0 2\pi}\int e^{-(\sigma_s^2-\sigma_w^2/2)\omega^2} d\omega$$ [2T]

$$= \frac{1}{N_0\sqrt{2\pi(2\sigma_s^2-\sigma_w^2)}} \cdot \frac{1}{\sqrt{2\pi\frac{1}{2\sigma_s^2-\sigma_w^2}}} \int e^{-\frac{\omega^2}{2\frac{1}{2\sigma_s^2-\sigma_w^2}}} d\omega$$

$$= \frac{1}{N_0\sqrt{2\pi(2\sigma_s^2-\sigma_w^2)}}$$ [3T]

4. Markov chains and martingales.

a) Denote the intensity by α (=0.1) and β (=0.2), respectively.

i) Let S (respectively T) denote the arrival time of the first white mouse (respectively black mouse). They have independent exponential distributions: [2A]

$$f(s) = \alpha e^{-\alpha s}, \quad s > 0$$
$$f(t) = \beta e^{-\beta t}, \quad t > 0$$

That the first mouse is black means

$$T < S$$

The probability is given by

$$\begin{aligned} P(T < S) &= \iint_{t<s} \alpha e^{-\alpha s} \beta e^{-\beta t} ds dt \\ &= \int_{s=-\infty}^{\infty} \alpha e^{-\alpha s} \int_{t=s}^{\infty} \beta e^{-\beta t}\, dt ds \\ &= \int_{s=-\infty}^{\infty} \alpha e^{-\alpha s} e^{-\beta s} dt \\ &= \frac{\alpha}{\alpha+\beta} = \frac{1}{3} \end{aligned}$$

[3A]

ii) Remember the Poisson process is memoryless, so

[2A]

$$\begin{aligned} P(T_1 < S_1, T_2 > S_2) &= \frac{\alpha}{\alpha+\beta}\frac{\beta}{\alpha+\beta} \\ &= \frac{\alpha\beta}{(\alpha+\beta)^2} \\ &= \frac{2}{9} \end{aligned}$$

[3A]

iii) Firstly, the total number of mice is a Poisson process with intensity 03.

[1A]

It can be thought of random selection of a Poisson process, which gives rise to two Poisson processes, each with arrival rate $\lambda = 0.15$. This yields two queues with difference service times. The number of mice is therefore the sum of the two queues, i.e., $k = k_1 + k_2$, where k_1 is Poisson distributed with parameter 0.15x10 = 1.5 and k_2 is Poisson distributed with parameter = 0.15x20 = 3.

[2A]

We know that the sum of two Poisson distributions is again Poisson with parameter equal to the sum of the individual distributions' parameters, i.e., 4.5. So the average number of mice is 4.5.

[2A]

b)

Fact: Being recurrent requires $\sum_n P\{X_{2n} = 0\} = \infty$, while being transient requires $\sum_n P\{X_{2n} = 0\} < \infty$.

i) Of the 2*n* steps, the chain must goes left for *n* steps, and right for *n*.
Therefore,

$$P\{X_{2n} = 0\} = \binom{2n}{n}\left(\tfrac{1}{3}\right)^n\left(\tfrac{2}{3}\right)^n \qquad \text{[4E]}$$

ii)

$$\sum_n P\{X_{2n} = 0\} = \sum_{n\geq 1}\left(\frac{1}{3}\right)^n\left(\frac{2}{3}\right)^n\binom{2n}{n} = \sum_{n\geq 1}\left(\frac{1}{3}\right)^n\left(\frac{2}{3}\right)^n\frac{(2n)!}{n!\,n!} \qquad \text{[3T]}$$

$$\approx \sum_{n\geq 1}\left(\frac{1}{3}\right)^n\left(\frac{2}{3}\right)^n 2^{2n}\frac{1}{\sqrt{\pi n}} = \sum_{n\geq 1}\left(\frac{8}{9}\right)^n\frac{1}{\sqrt{\pi n}}$$

$$< \sum_{n\geq 1}\left(\frac{8}{9}\right)^n = 9 \qquad \text{[3T]}$$

Since the sum converges, we conclude that the origin is a transient state.